

# Business Problem Statement

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A growing insurance provider, Prism Insurance Pvt. Ltd., wants to better understand its financial exposure, claim volumes, and customer retention rates in order to improve profitability and operational efficiency. The management team has noticed a significant imbalance between collected premiums and overall claim payouts. They are particularly interested in uncovering which factors—such as policy types (Travel, Health, Auto), demographic age groups, or current claim statuses—are driving the highest financial risk and influencing customer churn.

The main task was to analyze the company's internal insurance dataset to answer the following overarching business question:

*"How can the company leverage policy and claims data to track profitability, identify high-risk demographics, and optimize customer retention strategies?"*

## Deliverables

- Data Preparation & Modeling (Power Query/Excel):** Clean and transform the raw, unformatted Excel dataset for analysis. This includes resolving data corruption errors (e.g., unreadable dates), standardizing naming conventions, and creating conditional categories (such as 'Age Group' and 'Active/Inactive' status) to enable deeper demographic analysis.
- Data Analysis:** Organize the data to extract actionable insights regarding claim pipelines. Identify the volume of rejected versus settled claims and calculate the exact financial gap between total Premium Amounts and potential Coverage Amounts.
- Visualization & Insights (Power BI):** Build an interactive dashboard that highlights key financial metrics, customer distribution by policy type, and demographic trends, enabling stakeholders to make quick, data-driven decisions regarding pricing and risk.
- Report and Presentation:** Write a clear project summary detailing your key findings (e.g., the high financial cost associated with the 'Old' age group and the 56% inactive customer rate) alongside strategic business recommendations.
- Project Repository:** Include all raw data files, Power Query transformation documentation, and the final `.pbix` dashboard file in a well-structured repository for presentation and review.

# Insurance Claim & Portfolio Analysis

## 1. Project Overview

This project analyzes insurance portfolio health and claims processing efficiency using policyholder data. The objective is to uncover insights regarding financial risk exposure, profitability across policy types, demographic impacts on claim amounts, and customer retention metrics to guide strategic operational decisions for Prism Insurance Pvt. Ltd.

## 2. Dataset Summary

- Total Records:** ~10,000 Customers (5,003 Male, 5,001 Female)
- Total Variables:** 15 (Post-transformation)
- Key Features:**
  - Demographics:* CustomerID, Gender, Age, Age Group (Engineered)
  - Policy Details:* PolicyNumber, PolicyType, PolicyStartDate, PolicyEndDate, PremiumAmount, CoverageAmount, Active/Inactive Status (Engineered)
  - Claim Details:* ClaimNumber, ClaimDate, ClaimAmount, ClaimStatus

## 3. Data Preparation & Cleaning (Power Query)

Before visualization, the raw Excel dataset underwent extensive cleaning and transformation in Power Query to ensure data integrity and enable advanced analytics:

### Key Transformations Applied:

- Header Standardization:** Expanded abbreviated column names for clarity (e.g., PolicyNum to PolicyNumber, PremiumA to PremiumAmount).
- Data Type Correction & Error Handling:** Resolved formatting limitations from the raw Excel file (such as ##### errors in date columns) by defining strict Date types for PolicyStartDate and PolicyEndDate.
- Null Value Management:** Identified literal text strings of "NULL" in the ClaimDate and ClaimNumber columns (associated with rejected/unfiled claims) and isolated them from valid date calculations.
- Feature Engineering (Conditional Logic):**
  - Created an **Age Group** column (Young Adult, Adult, Old) by binning the numerical Age data to enable demographic risk profiling.
  - Created an **Active/Inactive** column by evaluating current dates against policy end dates to measure customer churn.

## 4. High-Level Financial Overview

The baseline financial health of the portfolio highlights a significant gap between collected revenue and claim payouts, measured against total potential exposure.

| Metric | Total Value (Millions) | Business Implication |
|--------|------------------------|----------------------|
|--------|------------------------|----------------------|

## 5. Data Analysis & Key Insights

### A. Revenue by Policy Type

Travel insurance is the dominant product line, generating more than double the premium revenue of the next highest category.

| Policy Type | Premium Amount |
|-------------|----------------|
| Travel      | 2.50 M         |
| Health      | 1.20 M         |
| Auto        | 1.00 M         |
| Life        | 0.70 M         |
| Home        | 0.60 M         |

### B. Risk Exposure by Age Group

The "Old" demographic represents the highest financial risk, driving the vast majority of claim payout amounts.

| Age Group   | Claim Amount |
|-------------|--------------|
| Old         | 8.90 M       |
| Adult       | 6.30 M       |
| Young Adult | 1.70 M       |

### C. Claims Processing Pipeline

An analysis of the claim statuses indicates a high friction point in claims approval. The volume of rejected claims outpaces settled claims, which may indicate issues with policy transparency, stringent underwriting rules, or high volumes of invalid claims being filed by customers.

- **Rejected Claims:** ~4,400 (Highest volume)
- **Settled Claims:** ~3,400
- **Pending Claims:** ~2,300 (Awaiting processing/investigation)

### D. Customer Retention & Churn

The newly engineered Active/Inactive metric reveals a critical vulnerability in customer retention. A majority of the portfolio consists of expired or cancelled policies.

| Status   | Customer Count | Percentage of Total |
|----------|----------------|---------------------|
| Inactive | 5,650          | 56.47%              |
| Active   | 4,360          | 43.53%              |

**Critical Insight:** The combination of a 56% churn rate (inactive policies) and total claim payouts (\$16.91M) vastly exceeding collected premiums (\$5.98M) suggests the current pricing and retention models are unsustainable and require immediate adjustment.

## 6. Business Recommendations

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Based on the data cleaning and subsequent analysis, we recommend the following strategic actions:

- **Premium Pricing Restructure:** With claim amounts (\$16.91M) exceeding premiums (\$5.98M), the company must urgently re-evaluate pricing models, particularly for the "Old" demographic which is driving \$8.9M in claims. Risk-adjusted premiums need to be increased for high-risk cohorts.
- **Retention & Renewal Campaigns:** With 56.47% of the customer base flagged as inactive, marketing teams should deploy targeted win-back campaigns offering loyalty discounts or bundled policies (e.g., combining highly popular Travel with lower-performing Home/Life).
- **Claims Process Audit:** Investigate the root cause behind the high volume of Rejected claims (4.4K). If customers misunderstand coverage terms, policy documentation should be simplified. If the rejection rate is due to fraud detection, the underwriting criteria should be tightened at the point of sale.
- **Capitalize on Travel Product:** Since Travel insurance brings in the highest premium revenue (\$2.5M), allocate more marketing budget to this segment to acquire new customers and offset losses in other departments.